

A STABILIZATION COUNTEREXAMPLE TO CONJECTURE 4.2 OF BOIX AND LIMA-PEREIRA

ABSTRACT. Boix and Lima-Pereira conjectured a Tor-length identity for complete-intersection ideals under the restriction $k \leq n - 1$. We prove a stabilization result that preserves the relevant Tor modules and colength, so imposing any fixed gap $n - k \geq c$ is equivalent to the unrestricted assertion. Stabilizing their Example 3.1 gives a minimal counterexample with $(n, k) = (4, 3)$: the Tor-length vector is $(4, 14, 14, 4)$, whereas the conjectured vector is $(4, 12, 12, 4)$. The example is nonreduced and does not address the motivating ICIS conjecture.

Let $(R, \mathfrak{m})$ be a regular local ring of dimension n , let I be generated by an R -regular sequence of length k , and let J be generated by an R -regular sequence of length n . Boix and Lima-Pereira conjectured that, when $k \leq n - 1$,

$$(1) \quad \ell_R\left(\mathrm{Tor}_i^R(R/I, R/J)\right) = \binom{k}{i} \ell_R(R/(I + J)), \quad 0 \leq i \leq k.$$

This is [1, Conjecture 4.2]; the unrestricted form $k \leq n$ was proposed in [2, Conjecture 4.9] and disproved in [1, Section 3] by examples with $k = n$.

For $c \geq 0$, let T_c denote the universal validity of (1) under the restriction $k \leq n - c$.

Theorem 1. *For every integer $c \geq 0$, the assertions T_c and T_0 are equivalent. Consequently, Conjecture 4.2 of [1] is false. Explicitly, let*

$$R = \mathbb{Q}[x, y, z, t]_{(x, y, z, t)},$$

$$I = (x^2, y^2, z^2), \quad J = (x^2 + xy, xz + z^2, y^2 + yz, t).$$

Then I and J are generated by regular sequences of lengths 3 and 4, and

$$\left(\ell_R\left(\mathrm{Tor}_i^R(R/I, R/J)\right)\right)_{i=0}^3 = (4, 14, 14, 4), \quad \ell_R(R/(I + J)) = 4.$$

Thus (1) predicts $(4, 12, 12, 4)$ and fails for $i = 1, 2$. Moreover, $(n, k) = (4, 3)$ is minimal in both coordinates.

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Proposition 2 (Stabilization). *For $r \geq 1$, set*

$$\begin{aligned} R_r &= R[t_1, \dots, t_r]_{\mathfrak{m}R[t_1, \dots, t_r] + (t_1, \dots, t_r)}, & I_r &= IR_r, \\ J_r &= JR_r + (t_1, \dots, t_r). \end{aligned}$$

Then R_r is regular local of dimension $n + r$, and I_r and J_r are generated by regular sequences of lengths k and $n + r$. For every i ,

$$\mathrm{Tor}_i^{R_r}(R_r/I_r, R_r/J_r) \cong \mathrm{Tor}_i^R(R/I, R/J), \quad R_r/(I_r + J_r) \cong R/(I + J),$$

and the corresponding lengths are equal.

Proof. The local map $R \rightarrow R_r$ is faithfully flat, so the chosen generators of I and J remain regular after extension. Moreover,

$$R_r/JR_r \cong (R/J)[t_1, \dots, t_r]_{(\mathfrak{m}/J, t_1, \dots, t_r)},$$

and $t_1, \dots, t_r$ is regular on this ring. This proves the assertions about regular sequences.

If $f_1, \dots, f_k$ generate I , their Koszul complex is a free resolution of R/I . Since $R_r/J_r \cong R/J$, there is an isomorphism of complexes

$$K_{R_r}(f_1, \dots, f_k) \otimes_{R_r} R_r/J_r \cong K_R(f_1, \dots, f_k) \otimes_R R/J.$$

Taking homology gives the Tor isomorphisms, and the quotient isomorphism is immediate. The modules on the left are annihilated by $(t_1, \dots, t_r)$, so length over R_r equals length over $R_r/(t_1, \dots, t_r) \cong R$. $\square$

Proof of Theorem 1. The case $c = 0$ is tautological. Let $c \geq 1$. The implication $\mathsf{T}_0 \Rightarrow \mathsf{T}_c$ is immediate. Conversely, assume T_c and take an arbitrary instance of T_0 in dimension n . Stabilizing with $r = c$ gives dimension $n + c$ and

$$k \leq n = (n + c) - c.$$

Thus T_c applies, and Proposition 2 transports (1) back to the original instance.

We now verify the displayed counterexample exactly. It is the one-variable stabilization of [1, Example 3.1]. Set

$$S = \mathbb{Q}[x, y, z], \quad J_0 = (x^2 + xy, xz + z^2, y^2 + yz), \quad A = S/J_0.$$

For graded reverse lexicographic order with $x > y > z$, a Gröbner basis of J_0 is

$$x^2 + xy, \quad xz + z^2, \quad y^2 + yz, \quad yz^2 - z^3, \quad z^4.$$

Hence A has homogeneous basis

$$1; \quad x, y, z; \quad xy, yz, z^2; \quad z^3.$$

In particular, $(x, y, z)^4 \subseteq J_0$, so J_0 is (x, y, z) -primary and its three generators form a regular sequence in $S_{(x, y, z)}$. The sequence x^2, y^2, z^2 is regular there as well.

In A , put $q_1 = x^2$, $q_2 = y^2$, and $q_3 = z^2$. The defining relations give

$$q_1 = -xy, \quad q_2 = -yz, \quad q_3 = z^2,$$

so q_1, q_2, q_3 form a basis of A_2 . Relative to this basis, the basis x, y, z of A_1 , and the basis z^3 of A_3 , multiplication $A_2 \otimes A_1 \rightarrow A_3$ is represented by

$$T = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{pmatrix}, \quad \det T = 4.$$

Thus this multiplication pairing is perfect.

Consider the Koszul complex $K(q_1, q_2, q_3; A)$. Its chain dimensions are 8, 24, 24, 8. Since the q_i have degree 2 and $A_d = 0$ for $d \geq 4$, only coefficient degrees 0 and 1 contribute to its differentials. Let $V = \mathbb{Q}^3$ with basis corresponding to the q_i . The two nonzero pieces of d_2 identify with

$$\bigwedge^2 V \longrightarrow V \otimes V, \quad u \wedge v \longmapsto u \otimes v - v \otimes u,$$

and, using the perfect pairing above, with

$$V^* \otimes \bigwedge^2 V \longrightarrow V, \quad \varphi \otimes (u \wedge v) \longmapsto \varphi(u)v - \varphi(v)u.$$

The first map is injective and the second is surjective, so both have rank 3. The degree-zero and degree-one parts of d_1 have ranks 3 and 1. Those of d_3 have ranks 1 and 3, the latter map being an isomorphism under the same perfect pairing. Hence

$$\text{rank } d_1 = 4, \quad \text{rank } d_2 = 6, \quad \text{rank } d_3 = 4.$$

Therefore the homology dimensions are

$$(8 - 4, 24 - 4 - 6, 24 - 6 - 4, 8 - 4) = (4, 14, 14, 4).$$

Since A is an Artinian local $\mathbb{Q}$ -algebra with residue field $\mathbb{Q}$, these dimensions are the corresponding lengths. Also, $A/(q_1, q_2, q_3)$ has basis $1, x, y, z$ (the q_i span A_2 and $z^3 = yq_3$), and hence length 4. Proposition 2 with $r = 1$ gives the stated four-dimensional example.

Finally, the identity holds for $k = 2$ by [1, Theorem 0.2(ii)] and for $k = 1$ by [1, Remark 4.3(ii)]. Every counterexample to Conjecture 4.2 therefore has $k \geq 3$ and $n \geq k + 1 \geq 4$, proving minimality. $\square$

Remark 3 (Scope). The complete intersection R/I is nonreduced, so the example lies outside the reduced ICIS setting of [1, Conjecture 4.1]. It also satisfies the inequalities proposed in [1, Question 5.1], since $(4, 14, 14, 4) \geq (4, 12, 12, 4)$ componentwise.

REFERENCES

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